CO2AT2 SCENARIO BASED QUESTIONS

```
1.#include <stdio.h>

int main() {
    float totalWaste = 5000;
    float recycledWaste, remainingWaste, percentage;

    printf("Enter amount of recycled waste (kg): ");
    scanf("%f", &recycledWaste);

    remainingWaste = totalWaste - recycledWaste;
    percentage = (recycledWaste / totalWaste) * 100;

    printf("\nTotal Waste: %.2f kg", totalWaste);
    printf("\nRecycled Waste: %.2f kg", recycledWaste);
    printf("\nRemaining Waste: %.2f kg", remainingWaste);
    printf("\nRecycled Percentage: %.2f%%", percentage);

    if (percentage >= 90)
        printf("\nPlant Efficiency: Excellent");
    else if (percentage >= 75)
        printf("\nPlant Efficiency: Good");
    else if (percentage >= 50)
        printf("\nPlant Efficiency: Average");
    else
        printf("\nPlant Efficiency: Poor");

    return 0;
}
```

```
2.#include <stdio.h>

int main() {
    int plan;
    float storageLimit, usedStorage, remainingStorage;
    float extraStorage, extraCharge, usagePercent;

    printf("Select Storage Plan:\n");
    printf("1. Basic - 100 GB\n");
    printf("2. Standard - 500 GB\n");
    printf("3. Enterprise - 2 TB (2048 GB)\n");
    printf("Enter choice: ");
    scanf("%d", &plan);

    if (plan == 1)
        storageLimit = 100;
    else if (plan == 2)
        storageLimit = 500;
    else if (plan == 3)
        storageLimit = 2048;
    else {
        printf("Invalid plan.\n");
        return 0;
    }

    printf("Enter used storage in GB: ");
    scanf("%f", &usedStorage);

    remainingStorage = storageLimit - usedStorage;
    usagePercent = (usedStorage / storageLimit) * 100;
```

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    printf("\nStorage Limit: %.2f GB", storageLimit);
    printf("\nUsed Storage: %.2f GB", usedStorage);
    printf("\nRemaining Storage: %.2f GB", remainingStorage);

    if (usedStorage > storageLimit) {
        extraStorage = usedStorage - storageLimit;
        extraCharge = extraStorage * 4;

        printf("\nStorage Status: Exceeded");
        printf("\nExtra Storage: %.2f GB", extraStorage);
        printf("\nExtra Charge: $%.2f", extraCharge);
    }
    else if (usagePercent >= 95) {
        printf("\nStorage Status: WARNING - Storage usage exceeds 95%%");
        printf("\nExtra Charge: $0.00");
    }
    else if (usagePercent >= 75) {
        printf("\nStorage Status: High Usage");
        printf("\nExtra Charge: $0.00");
    }
    else {
        printf("\nStorage Status: Normal");
        printf("\nExtra Charge: $0.00");
    }

    return 0;
}

3.#include <stdio.h>

int main() {
    int i;
    float quality, delay;
    float totalQuality = 0, deliveryCount = 0;
    int processedProjects = 0;
    char cancelled, stop;

    for (i = 1; i <= 20; i++) {
        printf("\nProject %d\n", i);

        printf("Is the project cancelled? (y/n): ");
        scanf(" %c", &cancelled);

        if (cancelled == 'y' || cancelled == 'Y') {
            printf("Project cancelled. Skipping...\n");
            continue;
        }

        printf("Enter Quality Score (0-100): ");
        scanf("%f", &quality);

        printf("Enter Delivery Delay (days): ");
        scanf("%f", &delay);

        totalQuality += quality;
        processedProjects++;

        /* Delivery percentage:
           Project is considered successfully delivered
           if delay is within 5 days. */
        if (delay <= 5)
            deliveryCount++;
```

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        /* Classification */
        if (quality >= 90 && delay <= 2) {
            printf("Performance: Excellent\n");
        }
        else if (quality >= 75 && quality <= 89 && delay <= 5) {
            printf("Performance: Good\n");
        }
        else if (quality >= 60 && quality <= 74) {
            printf("Performance: Average\n");
        }
        else if (quality < 60) {
            printf("Performance: Needs Improvement\n");
        }
        else {
            printf("Performance: Does not meet Good/Excellent criteria\n");
        }

        printf("Should management stop the evaluation? (y/n): ");
        scanf(" %c", &stop);

        if (stop == 'y' || stop == 'Y') {
            printf("Management stopped the evaluation.\n");
            break;
        }
    }

    if (processedProjects > 0) {
        printf("\n===== SUMMARY =====\n");
        printf("Projects Processed: %d\n", processedProjects);
        printf("Average Quality Score: %.2f\n",
               totalQuality / processedProjects);
        printf("Delivery Percentage: %.2f%%\n",
               (deliveryCount / processedProjects) * 100);
    }
    else {
        printf("\nNo projects were processed.\n");
    }

    return 0;
}
```

```
4.#include <stdio.h>

int main() {
    int day;
    int defectChips;
    int totalChips = 10000;
    float defectPercentage;
    int acceptedChips;
    char maintenance, breakdown;

    for (day = 1; day <= 30; day++) {

        printf("\n===== Day %d =====\n", day);

        printf("Is this a maintenance day? (y/n): ");
        scanf(" %c", &maintenance);

        /* Skip production on maintenance days */
        if (maintenance == 'y' || maintenance == 'Y') {
            printf("Maintenance day. Production skipped.\n");
            continue;
        }
```

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        printf("Did a machine breakdown occur? (y/n): ");
        scanf(" %c", &breakdown);

        /* Stop production if breakdown occurs */
        if (breakdown == 'y' || breakdown == 'Y') {
            printf("Machine breakdown occurred.\n");
            printf("Production evaluation stopped.\n");
            break;
        }

        printf("Enter number of defective chips: ");
        scanf("%d", &defectChips);

        if (defectChips < 0 || defectChips > totalChips) {
            printf("Invalid number of defective chips.\n");
            continue;
        }

        /* Calculate defect percentage */
        defectPercentage = ((float)defectChips / totalChips) * 100;

        /* Calculate accepted chips */
        acceptedChips = totalChips - defectChips;

        printf("Defect Percentage: %.2f%%\n", defectPercentage);
        printf("Accepted Chips: %d\n", acceptedChips);

        /* Assign quality grade */
        if (defectPercentage < 1) {
            printf("Quality Grade: A\n");
        }
        else if (defectPercentage <= 3) {
            printf("Quality Grade: B\n");
        }
        else if (defectPercentage <= 5) {
            printf("Quality Grade: C\n");
        }
        else {
            printf("Quality Grade: Reject Batch\n");
        }
    }

    printf("\n30-day production evaluation completed/stopped.\n");

    return 0;
}
```